FEDERAL DEMOCRATIC REPUBLIC OF ETHIOPIA • MINISTRY OF EDUCATION

ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE EXAMINATION (ESSLCE)

SCHOLASTIC APTITUDE TEST (SAT) — 2016 E.C. / 2024 G.C. | BOOKLET: 094 | CODE: 03

GENERAL DIRECTIONS:

This booklet contains 60 multiple choice questions. Section 1: Verbal Reasoning (1-35) & Section 2: Quantitative Reasoning (36-60). Time Allowed: 2 Hours (120 Minutes). Choose the single best alternative from A, B, C, or D.

[Air Pollution and Environmental Health]

Air pollution, one of the most pressing environmental concerns, is posing significant risks to human health and the planet's well-being. It is characterized by the presence of pollutants in the air. It's not just about breathing problems; it also harms the heart and makes life less enjoyable. Particulate matter, consisting of tiny particles suspended in the air, is a major contributor to air pollution. These particles can penetrate deep into the lungs, causing respiratory diseases and aggravating existing health conditions. Ground-level ozone, another common air pollutant, can lead to respiratory problems and the formation of smog.
air pollution, it is imperative to adopt sustainable practices and reduce dependence on fossil fuels. Governments play a crucial role in enforcing stringent emission standards for industries and promoting public transportation. Additionally, individuals can contribute by using energy-efficient
Tenomba

SECTION: READING COMPREHENSION

1. In the phrase "the planet's well-being", what does "the planet" refer to?

A. The earth
B. The sun
C. The moon
D. The nine planets

2. How can an air pollution-free environment be created for the new generation?

A. By planting trees every year
B. By restricting private use of car
C. By devising rules and regulations
D. Tackling the causes of air pollution from the source

3. As used in the passage, what is the meaning of the word "imperative"?

A. compulsory
B. tentative
C. conditional
D. optional

4. Which one of the following is NOT an air pollutant?

A. smog
B. carpooling
C. ground-level ozone
D. particulate matter

SECTION: ANTONYMS

5. DAUNTING

A. Discouraging
B. Intimidating
C. Frightening
D. Entertaining

6. IMMACULATE

A. Filthy
B. Spotless
C. Lucid
D. Tidy

7. RAVAGE

A. Devastate
B. Ruin
C. Damage
D. Renovate

[Population Dynamics and Sustainable Growth]

Population, which means the total number of people living in a certain area, is a critical aspect of our world. However, this simple number has far-reaching effects on our world. It greatly influences our societies, economies, and environment. When population grows too rapidly, it can create problems like shortages of food, clean water, and housing. On the flip side, a declining population can lead to workforce shortages and economic difficulties. Striking a balance is essential.
implement various strategies. They promote family planning, educate people about birth control, and create job opportunities to reduce the pressure on resources. Education is a key tool here as it empowers individuals to make informed decisions about family size.
understanding and managing population is crucial for ensuring a better quality of life for everyone
To manage population growth, governments and communities

In essence,

SECTION: READING COMPREHENSION

8. What does to "make informed decision" mean in the above passage? It means decision:

A. based on facts and information.
B. made for the sake of satisfying oneself.
C. made for the sake of satisfying others.
D. based on mere personal opinion.

9. What is the theme of the passage?

A. Population control and its disadvantages.
B. Decline in population size has problems.
C. Population growth always poses challenges.
D. The need to balance birth rate and death rate.

10. Based on the information given in the passage, which one of the following statements is correct?

A. Having the requisite understanding about population is what matters most.
B. Both the government and the community are responsible for controlling population.
C. The role of the community is highly constrained.
D. It is the sole responsibility of the government to control population growth.

11. What does the word "far-reaching" mean according to the context of the passage?

A. facilitating resource allocation
B. impacting economic development
C. affecting certain areas
D. going to faraway places

SECTION: ANALYTICAL & LOGICAL REASONING

12. Statement: Thomas Edison failed so many times before he invented the light bulb. Instead of quitting, however, Edison tried again and again until he finally achieved success. His achievement was applauded as an extraordinary determination.
Conclusion I: A problem can be a chance to do one's best.
advisable to have perseverance in the face of challenges.
- Conclusion II: It is
- Conclusion III: One has to persist in a problem whatever the consequences may be.
Which one of the above statement is:

A. Conclusions I and II
B. Conclusions II and III
C. Conclusion II
D. Conclusion I

13. It is apparent that most of the well-paid jobs in today's thriving economy require a higher level learning, usually a university degree. Conversely, many of the new jobs being created and flourishing, ranging from medium level businesses to huge manufacturing industries mainly require knowledge other than just having a degree. For workers in these new jobs, basic skills in communication, interpersonal skills and team work play a crucial role in getting a job and attaining career development.

A. Career development in most jobs is impossible without producing a degree.
B. A university degree is irrelevant for the newly created and flourishing jobs.
C. The new jobs that are being created rather require knowledge of communication skills.
D. Holding a university degree is crucial for career development.

14. - Statement I: Several sub-cities in Addis Ababa get one or two days of water supply per week.
Based on this information, it is valid to conclude that:
supply of water in the city.
- Statement II: There is a significant imbalance between demand for and
- Statement III: There is a booming water trading
Which one of the following ideas is TRUE

A. Each of the three is an independent cause for three different effects.
B. Statement III is the cause and statements I and II are its effects.
C. Statement I is the cause and statements II and III are its effects.
D. Statement II is the cause and statements I and III are its effects.

15. Statement: XYZ company workers who have their own cars are constantly complaining as they do not get parking spaces either inside or around the compound of the company. However, the company administration couldn't give any solutions to the complaints of the workers.
Conclusion II: The workers should leave their cars at home.
workers should rather sell their cars.
- Conclusion III: The
- Conclusion I: The workers should start using public transportation.
Which one of the following conclusions would

A. Conclusion I
B. Both conclusions I and II
C. Conclusion II
D. Conclusion I

16. Five students (namely Ha, Ga, Ta, Za, and Ya) all from the same grade level sat for four exams on four different subjects: English, History, Geography, and Mathematics.
difficult.
- For Ha, History was the difficult while Mathematics was the most
- For Ga, Geography the easiest what Ha found them to be (Math
- To Ta both History and Mathematics test, Geography hardest).
- To Za the easiest (History) was the most difficult.
- For Ya, what the most difficult one for most of the students?

A. Mathematics
B. English
C. History
D. Geography

According to the text, which

17. Assumption: Most traffic accidents in Ethiopia are caused mainly because of what many drivers commit while driving: such as talking on the phone, exchanging text messages, violation of traffic regulations, and recklessness among other causes.
Conclusion I: Most traffic accidents are caused because drivers do not have the requisite driving skills.
the decisions such drivers make while they are driving.
- Conclusion II: Most traffic accidents are caused because of the ill-behaviors of such drivers.
- Conclusion III: Most traffic accidents are caused because of the wrong
Which one of the following conclusions is more logical for the above assumption?

A. Both conclusion II and III
B. Both conclusions I and II
C. Conclusion II
D. Conclusion I

SECTION: WORD SUBSTITUTION

18. Risk factors such as smoking, overweight, lack of physical exercise and stress **exacerbate blood pressure, heart disease, and gastro-intestinal problems.**

A. moderate
B. mitigate
C. aggravate
D. improve

19. Unethical work habits **jeopardize individuals' development efforts and a country's overall growth potential.**

A. detriment
B. facilitate
C. speed-up
D. accelerate

20. Lowland areas usually get **intermittent rain; as a result, they have low animal and crop productions.**

A. substantial
B. frequent
C. regular
D. sporadic

SECTION: SENTENCE COMPLETION

21. Many health professionals believe that laughter has _____ effects on the human body as it enhances both physical and emotional health.

A. detrimental
B. harmful
C. therapeutic
D. adverse

22. The _____ forests in the countryside accompanied by the chirping of birds and the fresh breeze are really inspirational.

A. abandoned
B. deserted
C. tranquil
D. noisy

23. As the class contains students from diverse cultural backgrounds, the class is a highly _____ one.

A. invariable
B. consistent
C. homogeneous
D. heterogeneous

24. To fight some of the incurable diseases the world is facing currently, scientists are _____ biotechnology to develop effective drugs.

A. setting aside
B. utilizing
C. discounting
D. overlooking

SECTION: SENTENCE CORRECTION

25. Which one of the following sentences is grammatically correct?

A. In my opinion, we can visited our friend when we return from school.
B. In my opinion, we can visit our friend when we return back from school.
C. In my opinion, I think, we can visit our friend when we back from school.
D. In my opinion, we can visit our friend when we are back from school.

26. Which one of the following sentences is grammatically correct?

A. The dog that was riding a bicycle barked at me.
B. The dog, while it was riding a bicycle, barked at me.
C. While it was riding a bicycle the dog barked at me.
D. The dog barked at me while I was riding a bicycle.

27. Which one of the following sentences is grammatically correct?

A. What things are you enjoy doing in your spare time?
B. What are the things you enjoy doing in your spare time?
C. What things you are enjoy doing in your spare time?
D. What things you enjoy doing in your spare time?

28. Which one of the following sentences is grammatically correct?

A. The population of China is more great than India.
B. The population of China is greater than in India.
C. The population of China is greater than that of India.
D. The population of China is greater than India.

SECTION: SYNONYMS

29. CONDONE

A. Forgive
B. Complicate
C. Confuse
D. Discourage

30. ADHERE

A. Decline
B. Drop
C. Abandon
D. Obey

31. FEASIBLE

A. Unrealistic
B. Impractical
C. Achievable
D. Ambitious

SECTION: WORD ANALOGIES

32. BOTANY : PLANTS :: _____ : SOIL

A. Sociology
B. Pedagogy
C. Pedology
D. Biology

33. WONDERFUL : SATISFACTORY :: GIGANTIC : _____

A. Broad
B. Large
C. Comprehensive
D. Spacious

34. JUSTICE : FAIRNESS :: PREJUDICE : _____

A. Rational
B. Bias
C. Compassion
D. Just

35. TEAM : PLAYER :: CREW : _____

A. Musician
B. Member
C. Vocalist
D. Trainer

SECTION: LINEAR SYSTEMS & ALGEBRA

36. For what value(s) of k, does the system of linear equations 2x + 3y = 11 and kx + 6y = -3 have a unique solution?

A. k = 4
B. k != 4
C. k = -4
D. k != -4

37. If (8x^2 + 1) / (3x + 6) = x^2 / (x + 2) - 3, then what is the value of 1 / (9x + 1)?

A. -1/5
B. 1/20
C. -1/18
D. 1/18

38. The difference of two numbers is 10, and the sum of three times the larger number and twice the smaller number is 55. Then which one of the following is the value of the larger number?

A. 10
B. 15
C. 6
D. 5

39. What percent of 16 is 12?

A. 70%
B. 50%
C. 25%
D. 75%

40. One root of the quadratic equation x^2 + bx + c = 0 is 4 times the other root. If b = 10, what are the roots of the equation?

A. -2 and -8
B. 2 and 8
C. -1/2 and -2
D. 1/2 and 2

41. How many different four-letter words which start with a vowel can be formed from the word MATH if the letters cannot be used more than once?

A. 24
B. 18
C. 8
D. 6

42. A bag contains 15 balls, numbered 1 through 15. What is the probability of selecting a number that is odd or a multiple of 4?

A. 1/3
B. 11/15
C. 1/5
D. 8/15

43. Let Triangle ABC be an isosceles triangle with Angle C = 90. If the length of side AC is 4 cm, then what is the length of side AB in centimeters?

A. 4sqrt2
B. 8
C. 4
D. 2sqrt2

44. Which one of the following can NOT be the degree measure of each interior angle of a regular polygon?

A. 140
B. 105
C. 120
D. 135

45. A data of numbers is given: 0 < a < b < c < d < e < f (7 distinct strictly increasing positive numbers).
data

A. Quantity P is greater than Quantity Q
B. Quantity P is less than Quantity Q
C. Quantity P and Quantity Q are equal
D. The relationship cannot be determined

Quantity P: Mode of the data

46. Let x, y, z be the lengths of the sides of a triangle. Consider the following quantities:

Quantity Q: Median of the

A. Quantity P is greater than Quantity Q
B. Quantity P is less than Quantity Q
C. Quantity P and Quantity Q are equal
D. The relationship cannot be determined

47. If f(x) = 10^(2x - 1), then which one of the following is equal to f^-1(x)?

Quantity P: x + y + 3

A. log sqrtx + 1/2
B. log sqrtx - 1/2
C. log(2x - 1)
D. log x + 1/2

Quantity Q: z + 3

48. Which one of the following is NOT true about the graph of the function f(x) = (7/5)^x?

A. Its y-intercept is 1.
B. It is an increasing function.
C. Its range is the set of all real numbers.
D. Its domain is the set of all real numbers.

49. What are the values of y and z, respectively, in the sequence: -1/31, 1/63, y, z

-1, 1/3, -1/7, 1/15,

A. -1/31 and 1/255
B. -1/127 and 1/255
C. -1/127 and 1/227
D. 1/129 and -1/255

50. Which one of the following is the value of x in the sequence: 188, ...?

23, 34, 56, 89, x,

A. 156
B. 144
C. 111
D. 133

51. What is the rationalizing factor of the denominator of 1 / cbrt4?

A. cbrt2
B. 2
C. sqrt2
D. 3sqrt2

52. If 3x - 2 = 3 - 5x, then which one of the following is equal to 7 - 4x?

A. -9/2
B. 19/2
C. 9/2
D. 13/2

53. The greatest common factor (GCF) and least common multiple (LCM) of two numbers are 12 and 336, respectively. If one of the numbers is 84, then which one of the following is the other number?

A. 48
B. 96
C. 36
D. 72

54. How many numbers in the set {-8, -5, -3, -1, 0, 2, 5, 7, 8} satisfy the condition |5 - 3x| < 11?

A. 3
B. 6
C. 4
D. 5

55. A father is 32 years old and his son is 8 years old now. In how many years will the father be three times as old as his son?

A. 3
B. 5
C. 6
D. 4

[Annual Income of Five Schools]

Table: Annual Income of Five Schools (P, Q, R, S, and T) in thousands of Birr from various sources:

- School P: Tuition fees = 120, Term fees = 24, Donations = 54, Grants = 60, Miscellaneous = 12, Total Income = 270
- School Q: Tuition fees = 60, Term fees = 12, Donations = 21, Grants = 54, Miscellaneous = 3, Total Income = 150
- School R: Tuition fees = 210, Term fees = 45, Donations = 60, Grants = 42, Miscellaneous = 120, Total Income = 477
- School S: Tuition fees = 90, Term fees = 24, Donations = 51, Grants = 60, Miscellaneous = 15, Total Income = 240
- School T: Tuition fees = 120, Term fees = 30, Donations = 60, Grants = 55, Miscellaneous = 15, Total Income = 280

SECTION: DATA INTERPRETATION

56. Which school has the lowest percentage of income from term fees out of its total income?

A. S
B. P
C. R
D. Q

57. Which school has the lowest ratio of miscellaneous to term fees?

A. S
B. T
C. Q
D. R

58. What is the average of the income from tuition fees of the five schools?

A. 90
B. 120
C. 100
D. 210

59. Which school has the lowest income from donations and term fees combined?

A. S
B. R
C. Q
D. T

60. Which school has the highest percentage of income from grants out of its total income?

A. P
B. S
C. Q
D. T

VERIFIED ANSWER KEYS AND PEDAGOGICAL EXPLANATIONS

Question 1 [Correct Answer: A]

In the context of environmental science and air pollution, "the planet" refers to planet Earth, where human health and ecosystems are affected.

Question 2 [Correct Answer: D]

The text explicitly states: "By collectively addressing the root causes of air pollution, we can create a healthier environment for current and future generations."

Question 3 [Correct Answer: A]

"Imperative" means absolutely necessary, essential, or compulsory.

Question 4 [Correct Answer: B]

Carpooling is an action to reduce emissions (sharing rides), not an air pollutant itself. Smog, ozone, and particulate matter are pollutants mentioned in the text.

Question 5 [Correct Answer: D]

"Daunting" means intimidating, discouraging, or overwhelming. Its opposite in this context is "Entertaining" (inviting/pleasant).

Question 6 [Correct Answer: A]

"Immaculate" means perfectly clean, neat, or spotless. Its exact antonym is "Filthy".

Question 7 [Correct Answer: D]

"Ravage" means to cause severe damage, ruin, or destruction. Options A, B, and C are synonyms; "Renovate" (to restore/repair) is the antonym.

Question 8 [Correct Answer: A]

An "informed decision" is one derived from accurate knowledge, education, and objective facts rather than guesswork or preference.

Question 9 [Correct Answer: D]

The core theme centers on balancing population growth (balancing rates of growth vs resources / balancing population dynamics) so it is neither growing too rapidly nor declining.

Question 10 [Correct Answer: B]

Paragraph 2 explicitly states: "To manage population growth, governments and communities implement various strategies."

Question 11 [Correct Answer: B]

"Far-reaching effects" refers to deep, wide, and extensive impacts across societies, economies, and environments.

Question 12 [Correct Answer: A]

Both Conclusion I (seeing failure/problems as an opportunity to try hard) and Conclusion II (the value of perseverance/determination) logically follow from Edison's narrative. Conclusion III is reckless and illogical.

Question 13 [Correct Answer: C]

The passage emphasizes that for new jobs, interpersonal skills, communication, and teamwork (soft skills) play a crucial role beyond just a formal degree.

Question 14 [Correct Answer: D]

The structural imbalance between water supply and demand (Statement II) causes water rationing/shortages (Statement I) and creates a thriving market for commercial water vendors (Statement III).

Question 15 [Correct Answer: B]

If parking is completely unavailable and the administration provides no solution, practical alternative solutions for the commute are to take public transportation (I) or leave cars at home (II). Selling cars (III) is an extreme and invalid assumption.

Question 16 [Correct Answer: C]

Let's tally "most difficult" / "hardest" for each student:
History & Math
Ta, and Ya (3 students), making it the most difficult subject overall.
- Ha: Math
- Za: Geography
- Ga: History
- Ya: History
- Ta:

Question was answered most difficult / hardest by Ga,

Question 17 [Correct Answer: A]

Phone use, texting, breaking rules, and recklessness are behavioral choices (Ill-behavior, Conclusion II) and actions taken behind the wheel (Decisions while driving, Conclusion II), rather than a lack of basic technical driving ability (Conclusion I).

Question 18 [Correct Answer: C]

"Exacerbate" means to make a problem or disease worse; its exact synonym is "aggravate".

Question 19 [Correct Answer: A]

"Jeopardize" means to put at risk or damage/harm (cause detriment to).

Question 20 [Correct Answer: D]

"Intermittent" means occurring at irregular intervals; non-continuous. "Sporadic" is its closest synonym.

Question 21 [Correct Answer: C]

Enhancing physical and emotional health describes positive/healing effects, which corresponds to "therapeutic".

Question 22 [Correct Answer: C]

"Tranquil" (peaceful, calm) fits the serene context of fresh breezes and chirping birds in nature.

Question 23 [Correct Answer: D]

"Diverse" cultural backgrounds mean the group is composed of varied elements, which is defined as "heterogeneous".

Question 24 [Correct Answer: B]

Scientists apply or make use of (utilize) biotechnology to research solutions.

Question 25 [Correct Answer: D]

Option A incorrectly uses past tense "visited" after modal "can". Option B uses redundant expression "return back". Option C uses redundant phrase "In my opinion, I think". Option D is grammatically complete and accurate.

Question 26 [Correct Answer: D]

Options A, B, and C contain misplaced modifiers suggesting the dog was riding a bicycle. Option D correctly clarifies that the human was riding the bicycle when the dog barked.

Question 27 [Correct Answer: B]

Option B follows proper English question structure (WH-word + auxiliary verb + subject + verb).

Question 28 [Correct Answer: C]

When making comparisons, we must compare population to population (using pronoun phrase "that of India"), not population to a country directly.

Question 29 [Correct Answer: A]

"Condone" means to accept, overlook, or forgive wrongdoing.

Question 30 [Correct Answer: D]

"Adhere" (to rules/guidelines) means to comply with or obey.

Question 31 [Correct Answer: C]

"Feasible" means possible or practical to do easily; equivalent to "achievable".

Question 32 [Correct Answer: C]

Botany is the scientific study of plants. Pedology is the scientific study of soils.

Question 33 [Correct Answer: B]

Wonderful is an extreme degree of satisfactory (degree of intensity). Similarly, Gigantic is an extreme degree of Large.

Question 34 [Correct Answer: B]

Justice is characterized by fairness (synonymous core trait). Prejudice is characterized by bias.

Question 35 [Correct Answer: B]

A team is made up of individual players; a crew is made up of individual members (or crew members).

Question 36 [Correct Answer: B]

A 2 2 system has a unique solution if the determinant of the coefficient matrix is non-zero: det([2, 3; k, 6]) != 0 => (2)(6) - (3)(k) != 0 => 12 - 3k != 0 => 3k != 12 => k != 4.

Question 37 [Correct Answer: C]

Notice 3x + 6 = 3(x+2). Rewrite LHS: (3x^2 + 1) / (3(x + 2)) = (x^2) / (x + 2) - 3. Multiply both sides by 3(x + 2): 3x^2 + 1 = 3x^2 - 9(x + 2) => 3x^2 + 1 = 3x^2 - 9x - 18. Cancel 3x^2: 1 = -9x - 18 => 9x = -19 => 9x + 1 = -18. Therefore, 1 / (9x + 1) = 1 / (-18) = -1/18.

Question 38 [Correct Answer: B]

Let x be the larger number and y be the smaller number. x - y = 10 => y = x - 10. 3x + 2y = 55. Substitute y: 3x + 2(x - 10) = 55 => 3x + 2x - 20 = 55 => 5x = 75 => x = 15.

Question 39 [Correct Answer: D]

(12 / 16) 100% = (3 / 4) 100% = 75%.

Question 40 [Correct Answer: A]

Let roots be r and 4r. Sum of roots = -b = -10: r + 4r = -10 => 5r = -10 => r = -2. The roots are r = -2 and 4r = -8.

Question 41 [Correct Answer: D]

Letters available: {M, A, T, H} (total 4 letters, Vowel = {A}). 1st position (must be a vowel): 1 choice ('A'). 2nd position: 3 remaining letters. 3rd position: 2 remaining letters. 4th position: 1 remaining letter. Total arrangements = 1 3 2 1 = 6.

Question 42 [Correct Answer: B]

Sample space S = {1, 2, 3, ..., 15}, total N = 15. Odd numbers (O): {1, 3, 5, 7, 9, 11, 13, 15} (8 numbers). Multiples of 4 (M): {4, 8, 12} (3 numbers). Intersection (O M): (0 numbers, since multiples of 4 are all even). Total favorable outcomes = 8 + 3 = 11. Probability = 11/15.

Question 43 [Correct Answer: A]

Since Angle C = 90 and Triangle ABC is isosceles, AC = BC = 4 cm. The hypotenuse AB is: AB = sqrt(AC^2 + BC^2) = sqrt(4^2 + 4^2) = sqrt32 = 4sqrt2 cm.

Question 44 [Correct Answer: B]

Interior angle of a regular polygon is given by = (n - 2) 180 / n => Exterior angle E = 180 - = 360 / n.
360/60 = 6 (regular hexagon)
= 75 => n = 360/75 = 4.8 (not an integer!). Thus, 105 cannot be an interior angle.
- For 140: E = 40 => n = 360/40 = 9 (regular nonagon)
- For 135: E = 45 => n = 360/45 = 8 (regular octagon)
- For 120: E = 60 => n =
- For 105: E

Question 45 [Correct Answer: D]

Since all values are strictly distinct (a < b < c < d < e < f), no value repeats, so there is no unique mode (or mode is undefined/not uniquely determined). Hence, relationship cannot be determined.

Question 46 [Correct Answer: A]

By the Triangle Inequality Theorem, the sum of any two sides of a triangle is strictly greater than the third side: x + y > z. Adding 3 to both sides yields x + y + 3 > z + 3. Thus, Quantity P > Quantity Q.

Question 47 [Correct Answer: A]

Set y = 10^(2x - 1). Take log of both sides: log y = 2x - 1 => 2x = log y + 1 => x = (1/2)log y + 1/2 = log sqrty + 1/2. Swap variables: f^-1(x) = log sqrtx + 1/2.

Question 48 [Correct Answer: C]

For an exponential function f(x) = a^x where a = 7/5 > 1: Domain is (-,) (all real numbers) - True. Base > 1 => strictly increasing - True. f(0) = (7/5)^0 = 1 => y-intercept is 1 - True. Range is (0,) (strictly positive real numbers), NOT all real numbers - False.

Question 49 [Correct Answer: B]

Look at signs and denominators:
and z is positive.
-1/127 and z = 1/255.
- Signs alternate: -, +, -, +, -, +, -, +. Thus y is negative
- Denominators follow pattern 2^n - 1: 1, 3, 7, 15, 31, 63, 127, 255.

So y =

Question 50 [Correct Answer: D]

Examine differences between consecutive terms:
differences increase by multiples of 11 (11, 22, 33, 44, 55, ...).
44: x = 89 + 44 = 133.
34 - 23 = 11
Next difference should be
Check next term: 133 + 55 = 188 (Matches!)
56 - 34 = 22

Question 51 [Correct Answer: A]

cbrt(4) = cbrt(2^2). To make the denominator a rational integer (2), we multiply top and bottom by cbrt(2) = cbrt2, since cbrt(2^2) cbrt2 = cbrt(2) = 2.

Question 52 [Correct Answer: C]

Solve for x: 3x + 5x = 3 + 2 => 8x = 5 => x = 5/8. Substitute into 7 - 4x: 7 - 4(5/8) = 7 - 5/2 = (14 - 5)/2 = 9/2.

Question 53 [Correct Answer: A]

For any two positive integers A and B, GCF(A,B) LCM(A,B) = A B.
336) / 84 = 336 / 7 = 48.

Question 54 [Correct Answer: C]

Solve inequality: -11 < 5 - 3x < 11. Subtract 5: -16 < -3x < 6. Divide by -3 (flip inequality direction): -2 < x < 16/3 5.33.
(No)
elements: {-1, 0, 2, 5} => Total = 4 numbers.
Check set values in range (-2, 5.33):

Question 55 [Correct Answer: D]

Let t be the number of years.

Question 56 [Correct Answer: D]

Calculate percentage of Term fees over Total income for each school:
= P: 24 / 270 8.89%

Question 57 [Correct Answer: C]

Ratio = Miscellaneous / Term fees:
- P: 12 / 24 = 0.50

Question 58 [Correct Answer: B]

Average = (120 + 60 + 210 + 90 + 120) / 5 = 600 / 5 = 120 (in thousand Birr).

Question 59 [Correct Answer: C]

School Q has the lowest ratio (0.25).
Sum of (Donations + Term fees):
+ P: 120 / 45 2.67

Question 60 [Correct Answer: C]

Percentage of Grants out of Total income:
= P: 60 / 270 22.22%
- R: 60 + 45 = 105
- S: 51

- R: 42
School Q has the highest percentage